SAGAR PUBLIC SCHOOL

Integrated Programme • **Topic: Work, Power & Energy**

JEE MAIN Practice Sheet – 1

Name: _____ **Class / Sec:** _____ **Date:** _____

Time Allowed: 60 Mins • **Total Questions: 20** • 12 from JEE Main PYQs + 8 from Module

A Energy & Conservation (Q1–8) **B** Work–Energy Theorem (Q9–14) **C** Power (Q15–17) **D** Vertical Circular Motion (Q18–20)

SECTION A – Energy & Application of Energy Conservation (Q1–Q8)

Q1. *[PYQ · JEE Main 2025, 3 Apr Morning]*

A particle is released from height S above the surface of the earth. At certain height its kinetic energy is three times its potential energy. The height from the surface of the earth and the speed of the particle at that instant are respectively.

(A) $\frac{S}{4}, \frac{3gS}{2}$ (B) $\frac{S}{2}, \frac{3gS}{2}$

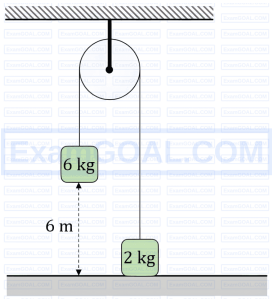
(C) $\frac{S}{4}, \sqrt{\frac{3gS}{2}}$ (D) $\frac{S}{2}, \sqrt{\frac{3gS}{2}}$

Q2. *[Module · Ex. S-1, Q28]*

The P.E. of a particle oscillating on x-axis is given as $U = 20 + (x-2)^2$; here U is in joules and x is in metres. Total mechanical energy of the particle is 36 J.
(i) Find the mean position. (ii) Find the maximum K.E. of the particle.

Q3. *[PYQ · JEE Main 2026, 6 Apr Evening]*

The velocity at which 6 kg mass (shown in figure) strikes the ground when it is released from a height of 6 m above the ground is _____ m/s. Assume pulley is massless and string is light and inextensible. (Take $g = 10\,\text{m/s}^2$)



(A) 7.74 (B) 7.20 (C) 6.55 (D) 4.50

Q4. *[Module · Ex. O-1, Q12]*

A block of mass m is hung vertically from an elastic thread of force constant mg/a. Initially the thread was at its natural length and the block is allowed to fall freely. The kinetic energy of the block when it passes through the equilibrium position will be:

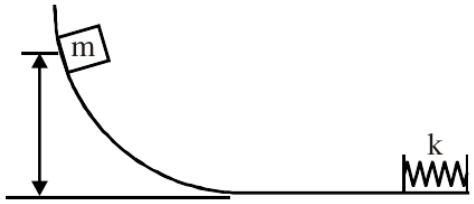
(A) mga (B) $mga/2$ (C) zero (D) $2mga$

Q5. *[PYQ · JEE Main 2021, 18 Mar Evening]*

A ball of mass 4 kg, moving with a velocity of $10\,\text{m s}^{-1}$, collides with a spring of length 8 m and force constant $100\,\text{N m}^{-1}$. The length of the compressed spring is x m. The value of x, to the nearest integer, is _____.

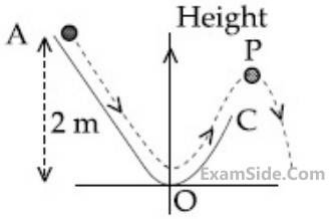
Q6. *[Module · Ex. S-1, Q23]*

A mass m slides from rest at height h down a smooth curved surface which becomes horizontal at zero height (see figure). A spring is fixed horizontally on the level part of the surface. The spring constant is k N/m. When the mass encounters the spring it compresses it by an amount $x = h/10$. If $m = 1\,\text{kg}$, $h = 5\,\text{m}$ then find $k/100$.



Q7. *[PYQ · JEE Main 2020, 7 Jan Morning]*

A particle ($m = 1\,\text{kg}$) slides down a frictionless track (AOC) starting from rest at a point A (height 2 m). After reaching C, the particle continues to move freely in air as a projectile. When it reaches its highest point P (height 1 m), the kinetic energy of the particle (in J) is: (Figure drawn is schematic and not to scale; take $g = 10\,\text{m s}^{-2}$)



Q8. *[Module · Ex. O-1, Q19]*

In a shotput event an athlete throws the shotput of mass 10 kg with an initial speed of 1 m/s at 45° from a height 1.5 m above ground. Assuming air resistance to be negligible and acceleration due to gravity to be $10\,\text{m/s}^2$, the kinetic energy of the shotput when it just reaches the ground will be

(A) 2.5 J (B) 5.0 J (C) 52.5 J (D) 155.0 J

SECTION B – Work–Energy Theorem (Q9–Q14)

Q9. *[PYQ · JEE Main 2022, 24 Jun Morning]*

A particle experiences a variable force $\vec{F} = (4x\,\hat{i} + 3y^2\,\hat{j})$ in a horizontal x-y plane. Assume distance in meters and force is newton. If the particle moves from point $(1, 2)$ to point $(2, 3)$ in the x-y plane, then Kinetic Energy changes by:

(A) 50.0 J (B) 12.5 J (C) 25.0 J (D) 0 J

Q10. *[Module · Ex. S-1, Q2]*

A body of mass 2 kg initially at rest moves under the action of an applied horizontal force of 7 N on a table with coefficient of kinetic friction = 0.1. Compute the
(a) work done by the applied force in 10 s,
(b) work done by friction in 10 s,
(c) work done by the net force on the body in 10 s,
(d) change in kinetic energy of the body in 10 s, and interpret your results.

Q11. *[PYQ · JEE Main 2022, 28 Jun Evening]*

A block of mass 2 kg moving on a horizontal surface with speed of $4\,\mathrm{m\,s^{-1}}$ enters a rough surface ranging from $x = 0.5$ m to $x = 1.5$ m. The retarding force in this range of rough surface is related to distance by $F = -kx$ where $k = 12\,\mathrm{N\,m^{-1}}$. The speed of the block as it just crosses the rough surface will be:

(A) zero (B) $1.5\,\mathrm{m\,s^{-1}}$
(C) $2.0\,\mathrm{m\,s^{-1}}$ (D) $2.5\,\mathrm{m\,s^{-1}}$

Q12. *[Module · Ex. O-1, Q8]*

A man who is running has half the kinetic energy of the boy of half his mass. The man speeds up by 1 m/s and then has the same kinetic energy as the boy. The original speed of the man was

(A) $\sqrt{2}$ m/s (B) $(\sqrt{2}-1)$ m/s
(C) 2 m/s (D) $(\sqrt{2}+1)$ m/s

Q13. *[PYQ · JEE Main 2026, 2 Apr Evening]*

A spherical ball of mass 2 kg falls from a height of 10 m and is brought to rest after penetrating 10 cm into sand. The average force exerted by sand on the ball is _____ N. (Take $g = 10\,\mathrm{m/s^2}$)

(A) 1980 (B) 2020 (C) 2000 (D) 1000

Q14. *[PYQ · JEE Main 2021, 22 Jul Evening]*

A porter lifts a heavy suitcase of mass 80 kg and at the destination lowers it down by a distance of 80 cm with a constant velocity. Calculate the work done by the porter in lowering the suitcase. (take $g = 9.8\,\mathrm{m\,s^{-2}}$)

(A) +627.2 J (B) −62720.0 J
(C) −627.2 J (D) 784.0 J

SECTION C – Power (Q15–Q17)

Q15. *[PYQ · JEE Main 2025, 7 Apr Morning]*

An object of mass 1000 g experiences a time dependent force $\vec{F} = \left(2t\,\hat{i} + 3t^2\,\hat{j}\right)$ N. The power generated by the force at time t is:

(A) $(2t^2 + 18t^3)$ W (B) $(3t^3 + 5t^5)$ W
(C) $(2t^3 + 3t^5)$ W (D) $(2t^2 + 3t^3)$ W

Q16. *[Module · Ex. S-1, Q17]*

Power applied to a particle varies with time as $P = (3t^2 - 2t + 1)$ watt, where t is in second. Find the change in its kinetic energy between time $t = 2$ s and $t = 4$ s.

Q17. *[PYQ · JEE Main 2020, 7 Jan Evening]*

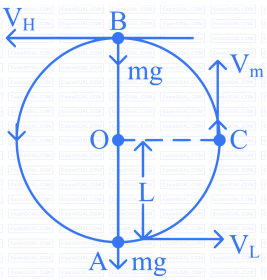
An elevator in a building can carry a maximum of 10 persons, with the average mass of each person being 68 kg. The mass of the elevator itself is 920 kg and it moves with a constant speed of 3 m/s. The frictional force opposing the motion is 6000 N. If the elevator is moving up with its full capacity, the power delivered by the motor to the elevator ($g = 10\,\mathrm{m/s^2}$) must be at least:

(A) 48000 W (B) 62360 W
(C) 56300 W (D) 66000 W

SECTION D – Vertical Circular Motion (Q18–Q20)

Q18. *[PYQ · JEE Main 2024, 29 Jan Evening]*

A bob of mass m is suspended by a light string of length L. It is imparted a minimum horizontal velocity at the lowest point A such that it just completes half circle reaching the top most position B. The ratio of kinetic energies $\dfrac{(K.E.)_A}{(K.E.)_B}$ is:



(A) 5 : 1 (B) 3 : 2 (C) 1 : 5 (D) 2 : 5

Q19. *[Module · Ex. S-1, Q32]*

A ball is attached to a horizontal cord of length L whose other end is fixed. (a) If the ball is released, what will be its speed at the lowest point of its path? (b) A peg is located a distance h directly below the point of attachment of the cord. If $h = 0.75\,L$, what will be the speed of the ball when it reaches the top of its circular path about the peg?

Q20. *[PYQ · JEE Main 2021, 25 Jul Morning]*

A pendulum bob has a speed of 3 m/s at its lowest position. The pendulum is 50 cm long. The speed of bob, when the length makes an angle of 60° to the vertical will be ($g = 10\,\mathrm{m/s^2}$) _____ m/s.

SELF ASSESSMENT (to be filled by the student after the test)

Time Taken: _____ min (Allowed: 60 min) **Attempted:** _____ / 20 **Unattempted:** _____

Correct: _____ **Incorrect:** _____ **Accuracy:** _____ %

Section	Questions	Correct	Incorrect	Topics / concepts I must revise
A – Energy & Conservation	8			
B – Work–Energy Theorem	6			
C – Power	3			
D – Vertical Circular Motion	3			

Personal Feedback (what went well / what went wrong / silly mistakes / plan for next sheet):

_____

_____

_____

_____

Student's Signature: _____ **Teacher's Remark / Signature:** _____